

CLASS X — PRE-BOARD EXAMINATION 2026

MATHEMATICS STANDARD (041) — PAPER 1

Date: 09 October 2026 | Time: 3 Hours | Maximum Marks: 80

GENERAL INSTRUCTIONS

1. This question paper contains 38 questions divided into Sections A to E.
2. Section A: $20 \times 1 = 20$; Section B: $5 \times 2 = 10$; Section C: $6 \times 3 = 18$; Section D: $4 \times 5 = 20$; Section E: $3 \times 4 = 12$ marks.

SECTION A — OBJECTIVE TYPE (20 MARKS)

- Q1.** If $\text{HCF}(96, 404) = 4$, then $\text{LCM}(96, 404)$ is: (A) 9696 (B) 9796 (C) 9690 (D) 9792
- Q2.** If α and β are zeroes of $2x^2 - 7x + 3$, then $\alpha^2\beta + \alpha\beta^2$ equals: (A) $\frac{21}{4}$ (B) $\frac{21}{2}$ (C) $\frac{7}{2}$ (D) $\frac{3}{2}$
- Q3.** The pair $3x + 2y = 5$ and $6x + ky = 10$ has infinitely many solutions when k is: (A) 2 (B) 4 (C) 6 (D) 8
- Q4.** If one root of $x^2 - 5x + k = 0$ is twice the other, then k equals: (A) $\frac{25}{9}$ (B) $\frac{50}{9}$ (C) $\frac{100}{9}$ (D) $\frac{10}{3}$
- Q5.** The 20th term of AP 7, 11, 15, ... is: (A) 79 (B) 83 (C) 87 (D) 91
- Q6-Q20.** [Midpoint formula, Trigonometric $\sec^2 - \cos^2$, Height & Distance 20m tower 60 deg, Circle tangent perpendicular to radius, Sector area 90 deg $r=14$, Cone curved surface area, Probability complement, Sphere volume $r=6$, Equal roots discriminant $b^2-4ac=0$.]

SECTION B & C — SHORT ANSWER (28 MARKS)

- Q21.** Euclid's division algorithm HCF of 867 and 255.
- Q22.** Find k for infinitely many solutions in linear system.
- Q23.** Sum of 15 terms of AP is 420 and $a=6$. Find common difference d .
- Q24.** Prove that $\sqrt{5}$ is irrational.
- Q25-Q31.** [Form quadratic polynomial from zeroes α^2, β^2 ; System of reducible equations $\frac{2}{(x+y)} + \frac{3}{(x-y)} = 5$; Triangle area & collinearity; Heights & Distances tower angles 30 deg & 60 deg; Quadrilateral circumscribing circle $AB+CD=AD+BC$.]

SECTION D & E — LONG ANSWER & CASE STUDIES (32 MARKS)

- Q32.** Speed of train word problem / Quadratic equation 360 km at uniform speed.
- Q33.** Basic Proportionality Theorem (Thales Theorem) proof and numerical application.
- Q34.** Solid hemisphere + cylinder surface area & volume calculation.
- Q35.** Statistics: Assumed mean method for marks distribution & median class identification.
- Q36. AP CASE STUDY:** Auditorium seating arrangement AP ($a=18, d=3, n=25$).

Q37. COORDINATE GEOMETRY CASE STUDY: Distance formula, midpoint & perpendicular bisector.

Q38. PROBABILITY CASE STUDY: Drawing balls without replacement (white, black, red).